

IoT - Enabled Temperature & Humidity Monitoring System

Submitted by: G. Sindhu Bhargavi

1. Introduction

This project demonstrates how to build a temperature and humidity monitoring system using virtual tools—**without any hardware cost**. By leveraging the Wokwi simulator and ThingSpeak cloud platform, all components (ESP32, DHT22 sensor) are simulated, and data is visualized online with live alerts for abnormal conditions.

2. Tools & Technologies

- **Wokwi Simulator:** Free online Web-based simulation for circuits (<https://wokwi.com>)
- **ESP32:** Virtual microcontroller (simulated in Wokwi)
- **DHT22 Sensor:** Measures temperature & humidity (simulated)
- **ThingSpeak:** Cloud IoT platform for data and charts (<https://thingspeak.com>)
- **Programming Language:** Arduino C++(in Wokwi IDE)
- **Web Browser:** Chrome/Edge/Firefox

3. Project Objectives

- **Monitor** temperature and humidity in real-time
- **Transmit** sensor readings to cloud (ThingSpeak)

- **Visualize** live data with automatic updating charts
- **Detect and alert** on abnormal temperature ($>35^{\circ}\text{C}$) or humidity ($>80\%$) values
- **Achieve all of the above with zero physical hardware**

4. System Overview

How it works:

1. ESP32 (simulated) reads DHT22 sensor data every 20 seconds
2. Sends readings and alert flags to ThingSpeak via HTTP
3. ThingSpeak dashboard graphs (fields 1 & 2) display temperature and humidity
4. Alert graphs (fields 3 & 4) display when thresholds are exceeded
5. Everything runs within a web browser; no hardware required

5. Implementation Steps

1. Set up Wokwi simulation:

- Added ESP32, DHT22
- Wired VCC, GND, and DATA connections

2. Program ESP32:

- Code to read sensors and detect alerts
- Sends four data fields to ThingSpeak:
 - Temperature ($^{\circ}\text{C}$), Humidity (%), Temp Alert (0/1), Humi Alert (0/1)

3. Configure ThingSpeak channel:

Created new channel with four fields matching the code

4. Run & Test:

- Started simulation and observed live data uploads
- Manually changed sensor values in Wokwi
- Verified that alerts appeared for out-of-range readings

6. Key Code Snippet

```
// Read DHT22 sensor
temperature = dhtSensor.getTempAndHumidity().temperature;
humidity = dhtSensor.getTempAndHumidity().humidity;

// Set alert flags
tempAlert = (temperature > 35.0);
humiAlert = (humidity > 80.0);

// Upload to ThingSpeak
String url = "http://api.thingspeak.com/update?api_key=XXXX";
url += "&field1=" + String(temperature);
url += "&field2=" + String(humidity);
url += "&field3=" + String(tempAlert ? 1 : 0);
url += "&field4=" + String(humiAlert ? 1 : 0);
```

➤ *See full code on my [GitHub repo](#).*

7. Key Results

- Live charts show both temperature and humidity updating every 20 seconds
- Alert status graphs spike when a threshold is crossed, then clear
- All data and alerts available online (can be accessed on any device/browser)
- System requires no physical hardware and is entirely free to use

8. Demonstration & Screenshots

Figure 1: Virtual Circuit Setup

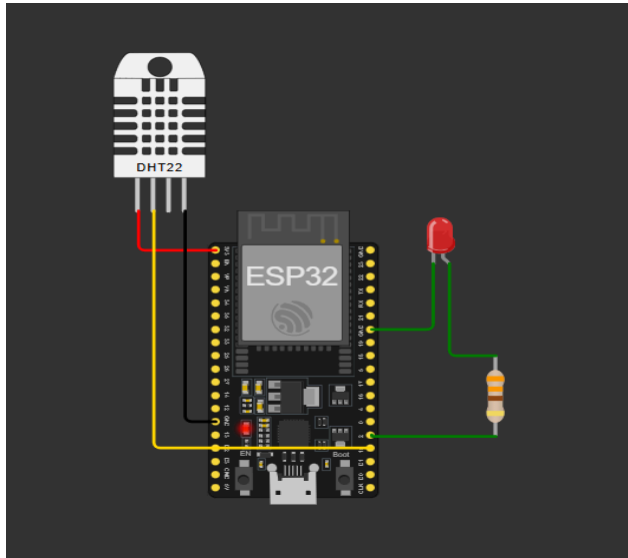


Figure 2: Serial Monitor - Normal Operation

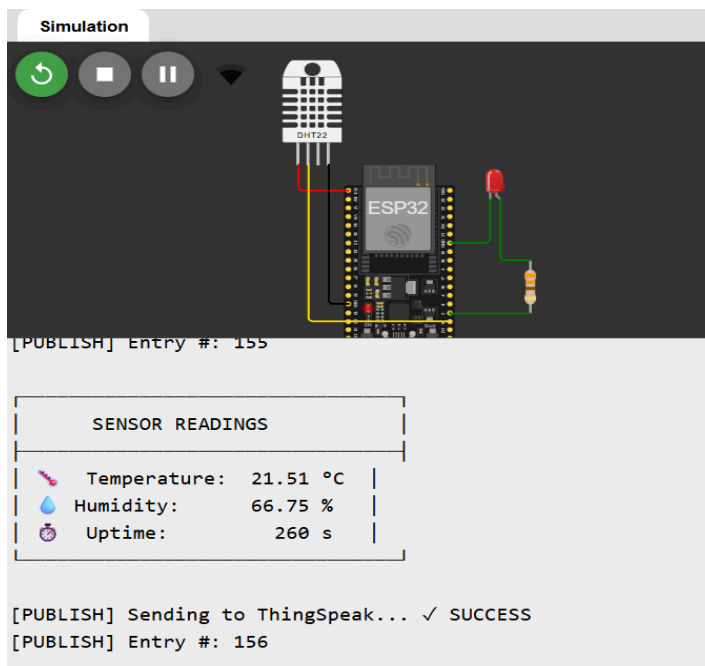


Figure 3: Serial Monitor - Alert Triggered

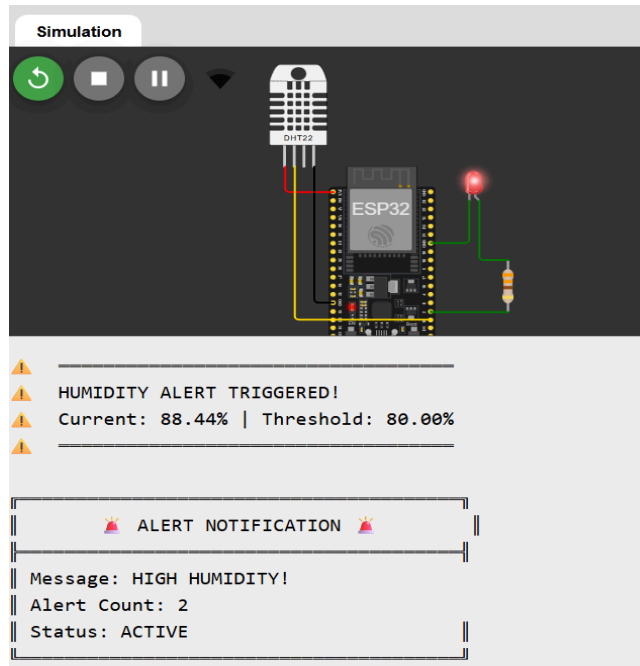


Figure 4: ThingSpeak Dashboard - Temperature & Humidity

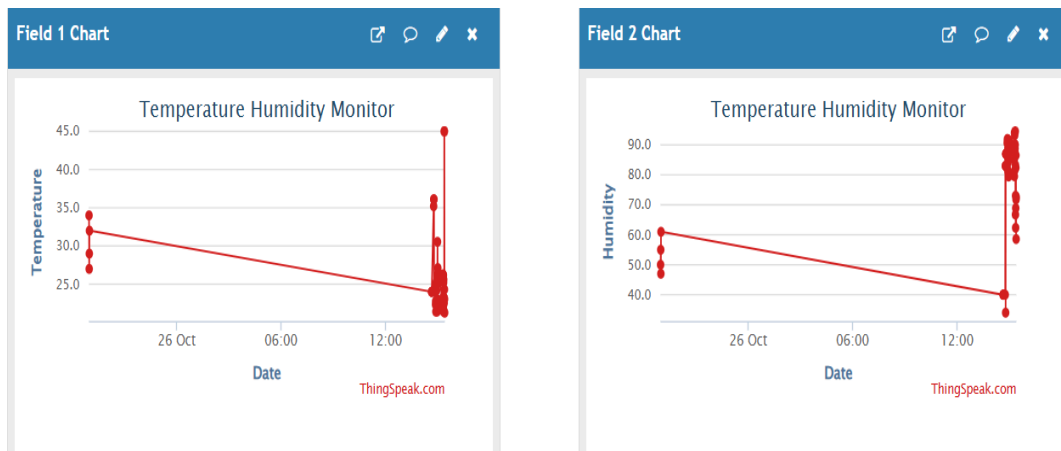
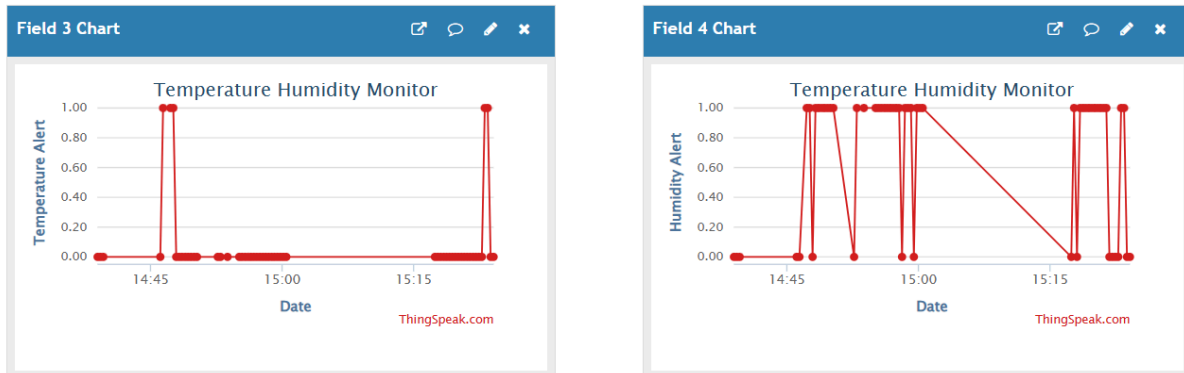


Figure 5: ThingSpeak Dashboard - Alert Status



9. Key Learnings

- How to build, test, and demonstrate an IoT system without hardware
- Sending data to the cloud and creating real-time graphs
- Automatic detection of “out of range” (alert) conditions
- Using professional coding and documentation practices, even in small projects

10. Future Improvements

- Add more sensors, e.g., for air quality, light, motion
- Enable email/SMS notifications using ThingSpeak’s apps or third-party services
- Deploy the same code to a real ESP32 when available

11. References

- [Wokwi Documentation](#)
- [ThingSpeak Documentation](#)
- [ESP32 Documentation](#)
- [DHT22 Datasheet \(Aosong AM2302\)](#)

11. Links

➤ **GitHub Repo:**

https://github.com/SindhuBhargavi1133/iot_monitoring_system